

Detection and Classification of Brain Tumor in MRI Images using Deep Convolutional Network

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Abstract— Brain tumor is a serious disease occurring in human being. Medical treatment process mainly depends on tumor types and its location. The final decision of neuro-specialists and radiologist for the tumor diagnosis mainly depend on evaluation of MRI (Magnetic Resonance Imaging) Images. The manual evaluation process is time-consuming and needs domain expertise to avoid human errors. To overcome this issue, Faster R-CNN deep learning algorithm was proposed for detecting the tumor and marking the area of their occurrence with Region Proposal Network (RPN). The selected MR image dataset consists of three primary brain tumors namely glioma, meningioma and pituitary. The proposed algorithm uses VGG-16 architecture as a base layer for both the region proposal network and the classifier network. Detection and classification results of the algorithm demonstrate that it is able to achieve an average precision of 75.18% for glioma, 89.45% for meningioma and 68.18% for pituitary tumor. As a performance measure, the algorithm achieved a mean average precision of 77.60% for all the classes.

Keywords— Brain tumor detection, glioma, meningioma, pituitary, Faster R-CNN, VGG-16, mean average precision, MRI.

I. INTRODUCTION

The abnormal growth of cells in human brain is called as brain tumor. A tumor which occurs in the brain or spinal cord is called as glioma and the tumor that arises from the meninges is called as meningioma. The abnormal cell growth in the pituitary gland is observed as pituitary tumor. The T1-weighted contrast enhanced Magnetic Resonance (MR) images can be used for the detection and localization of brain tumor. Based on color contrast differences, it is also able to differentiate brain tissues, edema and cerebrospinal fluid [1]. The early brain tumor detection is very much needed for effective treatment. For medical image diagnosis, the images can be obtained from various imaging modalities namely Positron Emission Tomography (PET), Magnetic Resonance Imaging (MRI) and Computed Tomography (CT). Among all these modalities, MRI is found to be best for brain tumor diagnosis. MRI is harmless because it is based on magnetic field and radio waves and do not pose any radiation hazard to human body [2].

Automatic tumor detection and classification methods are needed to reduce diagnosis time and overcome human errors before making any decisions. Mostly fundamental image segmentation techniques were applied to isolate brain tumor region. Deep learning algorithms were widely used for

detecting the visual objects in colour images. In this work, Faster R-CNN algorithm was chosen for detecting and classifying the brain tumor types. The Faster R-CNN model uses VGG-16 as base network. The algorithm highlights the tumor region and publishes the confidence scores along with boundary. The algorithm was trained to detect three primary brain tumors namely glioma, meningioma and pituitary.

This paper discusses about the literature survey in Section-II. Section-III describes about the image data preparation and the proposed Faster R-CNN model for the MR image dataset. The implementation of the algorithm is discussed in Section IV. The results and discussions are explained in Section V. Conclusion is given in Section VI.

II. LITERATURE REVIEW

Major difficulty in brain tumor detection and classification is the appearance of tumor region. The tumor region size, shape, location and intensity vary from image to image for a particular type. Usman Akram et. al [3], implemented a computer aided system for brain tumor detection, in which global thresholding technique was incorporated for segmenting the tumor. A. Hazra et. al [4], proposed a method for brain tumor detection and localization on MR images. It has three major steps namely pre-processing, edge detection (Sobel, Prewitt and Canny) and segmented the tumor region with k-means clustering algorithm.

Shahrjar et. al [5], proposed an automatic approach for brain tumor detection. The brain tumor images were pre-processed using image enhancement filters, further segmentation was performed and features were extracted for identifying the tumor. Deshmukh et. al [6], explained different brain tumor segmentation methods such as thresholding, region growing, K-means clustering, C-means clustering etc.

Parasuraman et. al [7], proposed a feed forward artificial neural network for classifying tumor and normal regions. The method consists of four different steps; first step deals with pre-processing using filters, the second step is image segmentation using clustering technique, the third step is feature extraction with gray-level co-occurrence matrix (GLCM) and the fourth step describes the classification of tumor using ensemble classifier. Sudiptha Roy et. al [8], implemented a symmetry analysis algorithm for classifying the tumor type and calculating the segmented area. Rajeswari et. al

[9], used Fast Fourier Transform (FFT) to identify the nature of the tumor type and its region. To analyze the signals converted from MR images, zero padded FFT and windowed FFT were used. It is found that the spectral leakage for the zero padded FFT is less than the windowed FFT. Madhukumar et. al [10], designed a method for quantitative comparison of fuzzy C-means and k-means segmentations with histogram guided initialization on tumor images. Mukesh Kumar et. al [11], proposed an automatic region growing method for tumor segmentation followed by texture analysis method to detect abnormality.

III. PROPOSED METHOD

An automatic brain tumor detection and classification method was implemented using Faster R-CNN algorithm. The VGG-16 architecture was chosen as a base network in Faster R-CNN for generating convolutional feature map to produce tumor region proposals followed by classification.

A. Image data preparation

The MR image dataset for the proposed work was constructed from the dataset of 233 patients collected by Jun Cheng et al. [12, 13]. The tumor image dataset is available online [14] along with its type and boundary information provided by radiologists. The image dataset is stored in **.mat** file MATLAB format. As a pre-processing step, each image in the dataset is normalized using min-max method [12, 13]. The image dataset consist of 2406 slices of T1-weighted contrast enhanced MR images of three classes namely glioma, meningioma and pituitary tumors as shown in Fig.1.

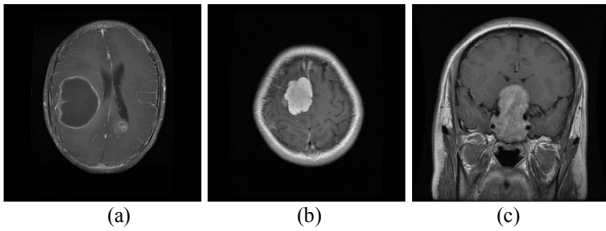


Fig. 1. Brain tumor types, (a) glioma (b) meningioma (c) pituitary tumor

The dataset consist of 805 slices of glioma, 694 slices of meningioma and 907 slices of pituitary tumors. After image normalization, all the images were stored in **.jpg** format and their respective annotations of tumor regions were stored in excel file. The annotation information comprises of bounding box information and tumor type. From the dataset, 80% of the

TABLE I. MR IMAGE DATASET PREPARATION

	Glioma	Meningioma	Pituitary tumors	Total No. of Samples
Training samples	638	551	735	1924
Testing samples	167	143	172	482

samples were chosen for training and remaining samples were taken for testing as shown in Table- I.

B. Faster R-CNN Architecture

The Faster R-CNN architecture [15] consist of three main blocks namely RPN, Region of Interest (RoI) pooling and Region based Convolutional Neural Network (R-CNN) for object detection. The input images were fed to base network, which is represented as tensors ($Height \times Width \times Depth$). The base network is made of VGG-16 model [16] up to fully connected layer. The convolutional feature map is obtained as output from the base network as shown in Fig.2. The dimensions of the convolutional feature map is represented as $Height/r \times Width/r$, where r is called as subsampling ratio. For VGG-16, ' r ' value is set to 16. As a first step, anchors (fixed bounding boxes) were placed in different scales; 256, 128 & 64 and aspect ratios of 1:1, 1:2 & 2:1 covering the whole image.

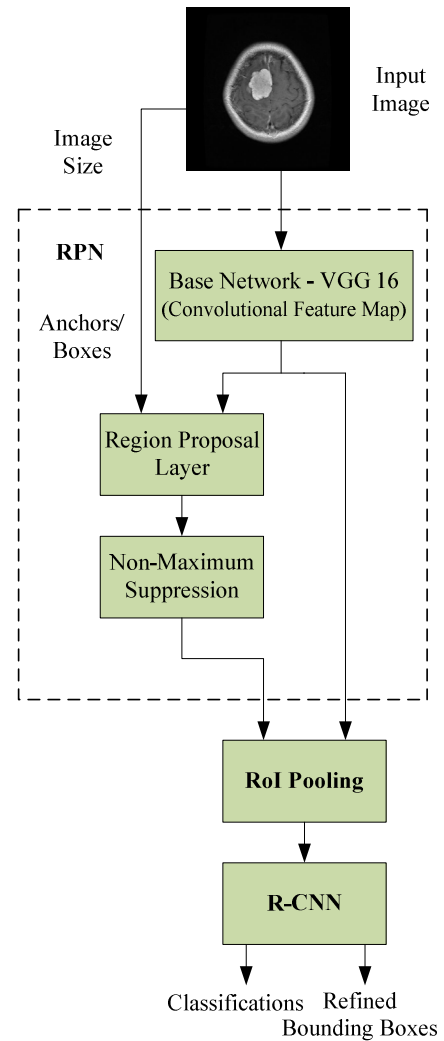


Fig. 2. Block diagram of the proposed Faster R-CNN algorithm

The region proposal layer takes all the anchors along with convolutional feature map and produces a set of good proposals with two different output values for each anchor

box. First output is the probability value predicting the presence of object inside the anchor boxes, which is called as confidence score. The confidence score is used to filter the undesirable predictions. The second output is the bounding box regression for adjusting the predicted anchors to exactly fit the object. The Non-Maximum Suppression (NMS) was used to avoid the duplicate proposals by setting Intersection over Union (IoU) threshold value between the proposals as 0.5.

The Region Proposal Network produces bunch of variable size rectangular object proposals for tumor detection. The generated proposals were fetched to the R-CNN algorithm to perform further classification and adjustment of the bounding box. For classifying each proposal, our aim is to crop the feature map regions and feed them to R-CNN which requires fixed-size input. The RoI Pooling layer crops the convolution feature maps and makes it to fixed size ($14 \times 14 \times 512$) using bilinear interpolation method. After cropping each proposal, max pooling is used to get a final feature map. Finally R-CNN classifies the proposals into one of the three classes and adjusts the bounding box of each proposal with respect to tumor region.

IV. IMPLEMENTATION DETAILS

Total of 2406 samples in the MR image dataset was distributed among training and testing phases. The training dataset consists of 1924 image samples and the testing dataset consists of 482 image samples as shown in the Table I. Training of Faster R-CNN algorithm was done using Stochastic Gradient Descent (SGD) optimizer with learning rate of 0.01 and momentum of 0.9. When compared with Batch Gradient Descent (BGD), the SGD performs only one update at a time during training and also it doesn't have redundant computations for larger datasets. Therefore it is much faster than the batch method [17]. The algorithm was trained for 120 epochs with 1000 iterations per epoch. The entire training process approximately took 16 hours on Nvidia Quadro P5000 GPU card with 16GB memory. The complete training process has four different losses namely RPN regression, RPN detection, R-CNN classifier and R-CNN box regression. After 120 epochs, the Faster R-CNN algorithm achieved a maximum training accuracy of 98.4%. The trained model was saved in hierarchical data format (.hdf5). All the losses and accuracy values of the algorithm along with elapsed time per each epoch were saved in excel file.

For testing, 167 samples of glioma, 143 samples of meningioma and 172 samples of pituitary tumors were chosen. During the testing process, the trained model was used to detect and classify these samples into three categories.

V. RESULTS AND DISCUSSION

The proposed Faster R-CNN algorithm uses VGG-16 as a base network for brain tumor detection and classification. The chosen MR image dataset consist of three classes namely glioma, meningioma and pituitary tumors. The tumor images belonging to the single class are not similar in terms of intensity and texture. Even though the images in the same class have different appearance, the proposed faster R-CNN

algorithm is able to efficiently detect and classify the tumor types.

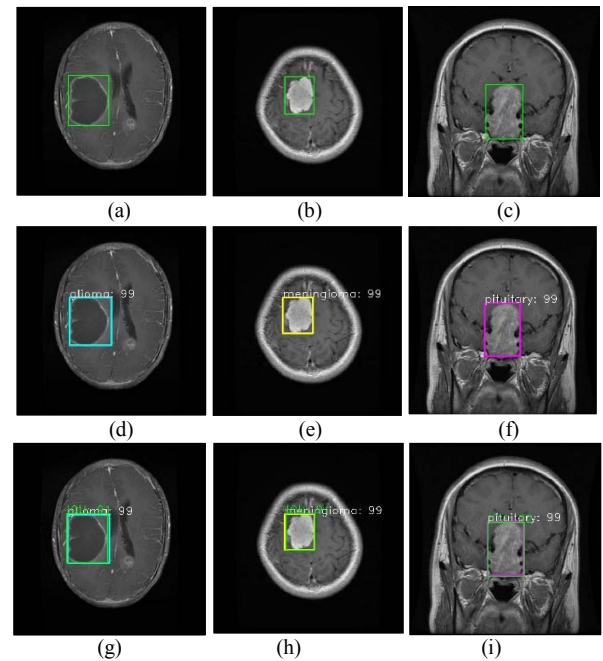


Fig. 3. Results of the proposed method; (a), (b), (c) are the ground-truth information representing the tumor region in glioma, meningioma and pituitary types. (d), (e) and (f) are the results of proposed Faster R-CNN algorithm for all the above types. (g), (h) and (i) shows the IoU calculation based on Faster R-CNN results.

The ground-truth boxes of input images Fig. 1(a), 1(b) & 1(c) are shown in Fig. 3(a), 3(b) & 3(c) respectively and it is used for calculating the IoU. The IoU is an evaluation metric used to measure the accuracy of an object detection method. The results of the proposed method for the input images were shown in Fig. 3(d), 3(e) & 3(f) respectively with bounding box, tumor type and confidence scores. Based on maximum confidence score, the proposed model selects a single bounding box for each image from the predicted boxes. The IoU was calculated using ground-truth information and bounding box prediction from the proposed model with the Eq.1. The IoU obtained for the input images based on Faster R-CNN results were shown in Fig. 3(g), 3(h) & 3(i).

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (1)$$


The mean Average Precision (mAP) is an important evaluation parameter for Faster R-CNN algorithm. To calculate mAP, the results of the proposed algorithm were recorded in excel sheet. For each individual tumor type, the results were separated and tabulated for the calculation of precision and recall values. Based on the confidence score, rows of the excel file were arranged in descending order. If the predicted and true classes are same with condition $IoU \geq 0.5$, then it is considered as True Positive (TP) otherwise it is

considered as False Positive (FP). The accumulated TP (Acc. TP) and accumulated FP (Acc. FP) are added in the excel file for all the rows. The equations for calculating precision and recall are given below.

$$Precision = \frac{Acc.TP}{Acc.TP + Acc.FP} \quad (2)$$

$$Recall = \frac{Acc.TP}{Total\ no.of\ boxes\ predicted} \quad (3)$$

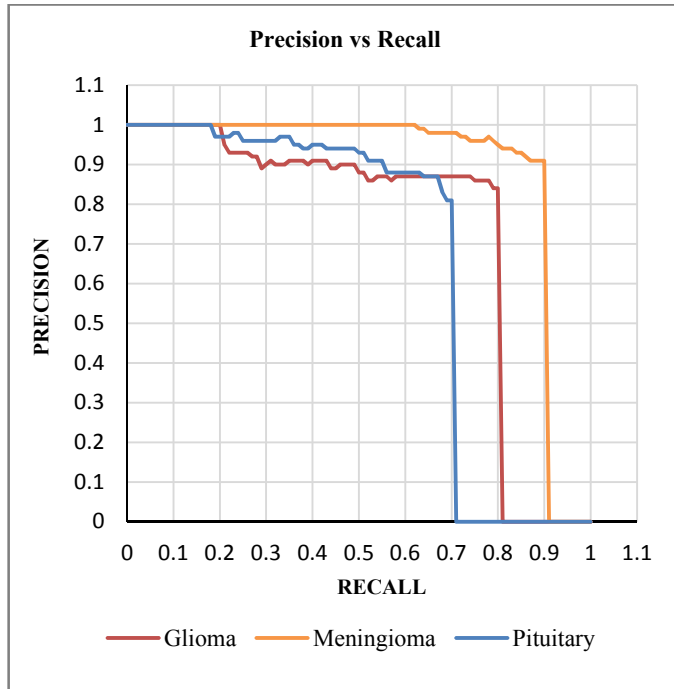


Fig 4. Precision vs Recall curves for glioma, meningioma and pituitary tumors

The maximum precision value is chosen when recall has the same values. The Precision vs Recall curves of glioma, meningioma and pituitary tumors were shown in Fig. 4. There

TABLE II. CALCULATION OF 11-POINT INTERPOLATED AVERAGE PRECISION FOR THREE TUMOR CLASSES REPRESENTING THE PRECISION-RECALL CURVES

Recall_11 points	Precision of glioma	Precision of meningioma	Precision of pituitary tumor
0	1	1	1
0.1	1	1	1
0.2	1	1	0.97
0.3	0.9	1	0.96
0.4	0.91	1	0.95
0.5	0.88	1	0.93
0.6	0.87	1	0.88
0.7	0.87	0.98	0.81
0.8	0.84	0.95	0
0.9	0	0.91	0
1	0	0	0

$$AP = \frac{1}{11} (Precision(Recall = 0) + \dots + Precision(Recall = 1)) \quad (4)$$

-after Average Precision (AP) was calculated by using the 11-point interpolation method [18]. From the Precision vs Recall curve, 11-points of Recall values were chosen between 0 and 1 with step size of 0.1 as shown in Table II and their respective precision values were used to calculate AP with Eq. 4. The AP for glioma, meningioma and pituitary tumors are 75.18%, 89.45% and 68.18% as shown in Table III. Overall mAP of 77.60% was obtained by taking mean value of all the three classes.

TABLE III. CALCULATION OF MEAN AVERAGE PRECISION FOR OVERALL PERFORMANCE MEASUREMENT

S.No.	Tumor Type	Average Precision (%)
1	Glioma	75.18
2	Meningioma	89.45
3	Pituitary	68.18
Mean Average Precision (%)		77.60

VI. CONCLUSION

This paper discusses about the automatic brain tumor detection and classification of MR Images using deep learning algorithm. The Faster R-CNN algorithm was chosen for detecting the tumor regions and classifying them in to three categories namely glioma, meningioma and pituitary tumor. For the Faster R-CNN algorithm implementation, a deep convolutional network architecture called VGG-16 was used as base network. The proposed algorithm efficiently identifies the brain tumor regions by choosing the optimal bounding box generated by RPN. A better mAP has been achieved for the detecting the brain tumor using the test dataset. This work can also be extended for calculating the percentage area of the tumor with respect to brain region. This method can also be generalized to other medical applications, such as skin lesion segmentation and classification.

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